Rajat M

✉️ rajat16127@gmail.com | 📞 +91 8700048301 | 📍 Mumbai, Maharashtra
linkedin.com/in/rajat16127 | github.com/rajat16127

PROFESSIONAL SUMMARY

- **Computer Vision & ML Engineer** with hands-on experience designing, training, and deploying real-time deep learning pipelines and YOLO-based edge detection systems.
- Demonstrated track record delivering end-to-end industrial vision solutions: automated frame extraction, annotation curation, custom model training, spatial event logic, and Dockerized deployment.
- Proficient in Python, OpenCV, Ultralytics YOLO, PyTorch/TensorFlow, and containerization workflows for production factory environments.
- Strong academic foundation in digital image processing, CNN architectures (transfer learning, feature maps), and data-centric performance optimization.

TECHNICAL SKILLS

- **Computer Vision:** Object Detection (YOLOv8/YOLOv11), Image Classification, Transfer Learning, OpenCV, Feature Extraction, Virtual Tripwire / Line Crossing Logic, Image Augmentation
- **Deep Learning & ML:** CNNs, Xception, PyTorch, TensorFlow, Keras, scikit-learn, Model Evaluation (mAP, IoU, Confusion Matrix, Precision-Recall)
- **Industrial Vision & MLOps:** Docker, Docker Hub container registries, Edge Inference Pipelines, OpenVINO model optimization, Custom Dataset Annotation & Cleaning, Video Stream Processing (RTSP/CCTV)
- **Programming & Tools:** Python (NumPy, Pandas, Matplotlib), Git, Linux/Bash, REST APIs, JSON data pipelines

WORK EXPERIENCE

Computer Vision Engineer — BitVivid Solutions

Feb 2026 – Present

- Engineered end-to-end industrial video analytics pipelines from raw RTSP factory footage to production inference on edge/on-premise servers.
- Curated custom domain-specific datasets: extracted keyframes, applied geometric/photometric augmentations, and eliminated edge-case labeling artifacts to boost object detection accuracy.
- Fine-tuned and optimized multiple Ultralytics YOLO architectures for manufacturing environments, minimizing false positives in variable lighting and occlusions.
- Programmed programmatic spatial tracking and event logic (virtual boundary crossing, line-based tallying, automatic snapshot triggers, and structured JSON telemetry export).
- Packaged inference microservices into lightweight Docker images, published container builds to Docker Hub, and accelerated CPU runtime using OpenVINO for seamless plant deployment.

COMPUTER VISION & DEEP LEARNING PROJECTS

Industrial Cotton Bale Detection & Conveyor Counting System | BitVivid Industrial Deployment | Project Report

Designed and delivered an end-to-end industrial computer vision solution for an automated cotton factory in Andhra Pradesh.

- **Pipeline & Model Training:** Sampled high-variance image frames from raw factory CCTV feeds, curated annotated training sets, and fine-tuned a custom YOLO object detector to recognize fast-moving cotton bales under shifting warehouse illumination.
- **Virtual Counting Logic:** Implemented a directional virtual tripwire across the conveyor belt coordinate space to reliably track bale centroids, assign unique entity IDs, and increment counts without duplicate tallies.
- **Metadata & Snapshot Engine:** Built an automated export service that generates high-resolution cropped audit snapshots alongside structured metadata files (camera ID, timestamp, bale sequence number, centroid coordinates) in JSON format upon line crossing.
- **MLOps & Deployment:** Containerized the complete inference engine and tracking logic using Docker, published production images to Docker Hub, and ensured turnkey deployment on target factory compute hardware.
- **Tech:** Python, YOLO (Ultralytics), OpenCV, Docker, Docker Hub, Linux

Deep Learning Fashion Image Classifier | Project Report

Engineered a multi-class image recognition pipeline classifying apparel across 10 distinct categories using transfer learning and deep convolutional networks.

- Leveraged a pretrained 71-layer **Xception** backbone, freezing base feature extractors and training custom dense heads with `GlobalAveragePooling2D`, dropout regularization, and softmax classification.
- Constructed an aggressive data augmentation pipeline via `ImageDataGenerator` (rotations, spatial shifts, shears, zooms, and horizontal flips) to enforce invariance against viewpoint shifts.
- Evaluated performance across precision, recall, and class-level confusion matrices, diagnosing false positive trends between visually similar classes.
- **Tech:** Python, TensorFlow, Keras, OpenCV, Matplotlib, NumPy

High-Dimensional Risk Classification & Preprocessing Engine | Project Report

Built scalable ML classification pipelines across 307k+ instances and 120+ features, handling extreme class imbalance and data sanitization.

- Engineered automated preprocessing pipelines incorporating median/mode imputation, multi-class categorical encoding, and feature scaling.
- Benchmarked tree-based ensembles (Random Forest) against linear baselines, tuning threshold boundaries to maximize recall and ROC-AUC scores.
- **Tech:** Python, scikit-learn, Pandas, NumPy

EDUCATION

Bachelor of Engineering in Information Technology — University of Mumbai

- **Relevant Coursework:** Image Processing, Artificial Intelligence, Machine Learning, Data Structures & Algorithms, Distributed Systems.

ADDITIONAL INFORMATION

- **Languages:** English, Hindi, Marathi
- **Focus Areas:** Industrial Computer Vision, Real-Time Edge Analytics, Object Detection & Tracking, Model Deployment